Credit Card Fraud Detection

Problem Statement

- Creditcard fraud is a majorchallenge in thedigital payment industry, causing significant financial losses to banks, merchants, and customers every year. As online transactions continue to increase, fraudsters use sophisticated techniques to perform unauthorized transactions and exploit security vulnerabili
- Key Points:
- Increasing number of online transactions leads to higher fraud risk.
- Fraudulent transactions are rare and difficult to detect.
- Highly imbalanced data affects model performance.
- Traditional methods are less effective for real-time detection.

Objectives:

- To analyze credit card transaction data and identify suspicious patterns.
- To preprocess and balance the dataset using SMOTE for improved model performance.
- To perform feature engineering by creating meaningful fraud-related features.
- To build and train an XGBoost model for fraud detection.
- To predict the probability of fraud for incoming transactions.
- To classify transactions as legitimate or fraudulent with high accuracy.
- To generate risk levels and recommendations based on prediction results.
- To develop a user-friendly web application using Flask for real-time fraud monitoring.
- To improve fraud detection efficiency while minimizing false alarms.

Methodology

- The proposed system follows a structured machinelearning pipeline to detect transactions accurately and efficiently
- **Steps Involved:**
- **Data Collection:**Obtain credit card transaction data from the dataset.Identify relevant features such as transaction amount, time, customer information, and fraud labels.
- **Data Preprocessing:**Handlemissingvalues andduplicate records.Normalizeand transformdatafor better model.
- **Data Balancing using SMOTE:**Address class imbalance by generating synthetic fraud samples.Ensure balanced representation of fraudulent and legitimate transaction.

System Architecture

The Credit Card FraudDetectionSystemisdesigned to processtransactiondata, extractmeaningful features, and predict fraudulent activities in real time using the XGBoost machine learning model. The system integrates data preprocessing, feature engineering, model prediction, and a Flask-based web interface to provide instant fraud detection and risk assessment.

Architecture Components.

Architecture Components:

1. Transaction Input Layer:

- **Receives transaction details from users.**
- **Inputs include transaction amount, time, location, device information, and merchant details.**

2. Data Preprocessing Layer

- **Cleans and validates input data.**
- **Handles missing values and transforms data into a suitable format.**

3. Feature Engineering Layer

- **Generates additional risk-related features such as:**
- **High Transaction**

Implementation

Theimplementationof the Credit CardFraud Detection System involves data preprocessing, feature engineering, model training, and deployment through a Flask web application. The XGBoost algorithm is used to classify transactions as legitimate or fraudulent based on transaction patterns and risk indication.

Implementation Steps:

1. Data Preparation: Load the credit card transaction dataset.Clean and preprocess the data.Handle missing values and duplicate records.

2. Data Balancing: Improve the model's ability to detect rare fraud cases.

3. Feature Engineering: Create additional features such as:

- High Transaction Amount
- Night Transaction
- Transaction Velocity

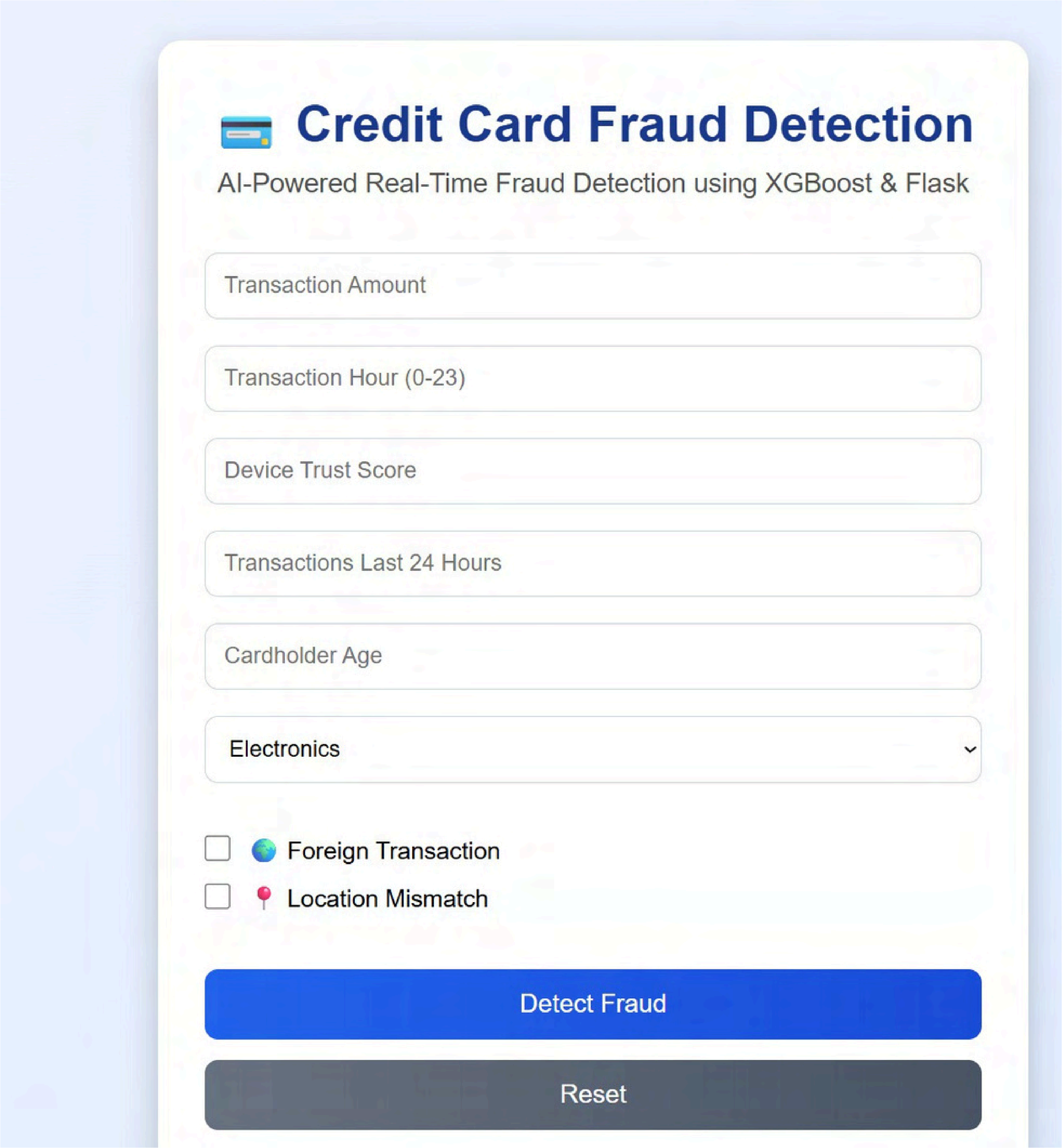
4. Model Development: Train the XGBoost Classifier using the processed dataset.Optimize model parameters for better accuracy and recall.

Results

The Credit Card Fraud Detection System successfully identified fraudulent transactions with high accuracy using the XGBoost machine learning model. The application was able to process transaction details in real time and provide instant fraud predictions through the Flask web interface.

After applying data preprocessing, SMOTE balancing, and feature engineering techniques, the model demonstrated improved performance in detecting fraudulent transactions while reducing false predictions. The generated fraud probability scores and risk levels helped classify transactions effectively into legitimate or fraudulent categories.

Key Results



Credit Card Fraud Detection
AI-Powered Real-Time Fraud Detection using XGBoost & Flask
Transaction Amount
Transaction Hour (0-23)
Device Trust Score
Transactions Last 24 Hours
Cardholder Age
Electronics
Foreign Transaction
Location Mismatch
Detect Fraud
Reset

✅ LEGITIMATE TRANSACTION

Fraud Probability

0.01%

Risk Level

LOW 🟢

Recommendation

✓ Transaction approved. No suspicious activity detected.

Transaction Details

💰 **Amount :** ₹ 10000

🕒 **Transaction Hour :** 10

🏬 **Merchant Category :** Electronics

🌏 **Foreign Transaction :** Yes

📍 **Location Mismatch :** No

📱 **Device Trust Score :** 100

⚡ **Transactions Last 24 Hours :** 2

👤 **Cardholder Age :** 25

Detection Information

🕒 **Detection Time :** 25-06-2026 07:47:19 PM

🤖 **Model :** XGBoost

⚠ FRAUD DETECTED

Fraud Probability

99.07%

Risk Level

HIGH 🔴

Recommendation

❌ Block the transaction immediately and notify the customer.

Transaction Details

💰 **Amount :** ₹ 1000000

🕒 **Transaction Hour :** 1

🏪 **Merchant Category :** Food

🌏 **Foreign Transaction :** Yes

📍 **Location Mismatch :** Yes

📱 **Device Trust Score :** 90

⚡ **Transactions Last 24 Hours :** 2

👤 **Cardholder Age :** 30

Detection Information

🕒 **Detection Time :** 25-06-2026 07:44:28 PM

🤖 **Model :** XGBoost

Future Scope

TheCreditCard Fraud Detection Systemcanbe furtherenhancedby incorporating advanced technologies and larger datasets to improve detection accuracy and scalability. Future developments can focus on real-time deployment, adaptive learning, and integration with modern financial systems.

Future Enhancements:

DeepLearning Models: ImplementadvancedmodelssuchasAutoencoders,LSTM,and NeuralNetworksfor improved fraud detection.

Real-Time Streaming Analysis: Integrate with payment gateways and banking systems for live transaction monitoring.

Adaptive Learning: Continuously retrain the model using new transaction data to detect emerging fraud patterns.

CloudDeployment: Deploy theapplicationoncloud platformsfor better scalability andaccessibility.

Analytics Dashboard: Add advanced visualizations and reporting features for fraud trend analysis and model monitoring.

TEAM MEMBERS

Credit Card Fraud Detection System

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2 Somepalli Madhuri

3 Bodapati Venkat Rao

4 Challa Hari Krishna

5 Pitta Gagan Kumar

6 Kuriti Rohith

7 Davaleswarapu V. N. S. Ashrita

MENTOR

Dr. Kanthi Kiran

We thank our mentor for their guidance and support.

Thank You